

PARK RANGER AI ASSISTANT



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1. EXECUTIVE SUMMARY

The Park Ranger AI Assistant project represents the culmination of a multi-phase technical initiative to design, build, and deploy a highly specialized Retrieval-Augmented Generation (RAG) system. This is an educational project built to demonstrate hybrid RAG techniques using a curated sample of park data from 10 Canadian parks and 10 U.S. parks.

Reliability and hallucination mitigation are central to the system. High-confidence factual inquiries are routed through a Q&A dataset, while broader or more nuanced queries fall back to dense vector search powered by ChromaDB and Sentence Transformers.

This report summarizes the full development lifecycle of the application, from raw DOM scraping and data sanitization to architecture design, embedding strategy, and final UI deployment via Streamlit. It serves as both a technical blueprint and a quality-assurance review of an end-to-end RAG pipeline.

2. DOMAIN SELECTION

The chosen domain for this RAG implementation is North American National Parks, specifically focusing on data aggregated from Parks Canada and the United States National Park Service (NPS). The selection of this domain was strategic, driven by several technical and data-quality factors.

High-Quality, Structured Documentation

Government park websites provide a wealth of highly structured, rigorously maintained documentation. Content is naturally partitioned into logical categories (e.g., Trail Guides, Camping Fees, Permit Regulations, Wildlife Safety). This clear hierarchical structure is an ideal ground for "structure-aware" chunking algorithms. Unlike unstructured social media or forum data, park documentation allows for highly predictable text extraction, reducing noise during the embedding process.

Cross-Border Contextual Complexity

Integrating data from both Parks Canada and the US NPS introduces a deliberate layer of complexity to test the LLM's retrieval accuracy. The system must reliably differentiate between similar entities across borders (e.g., Glacier National Park in Canada vs. Glacier National Park in the US) and different regulatory frameworks (e.g., Discovery Pass in Canada vs. America the Beautiful Pass in the US). This prevents the AI from relying purely on semantic similarity and forces it to synthesize geographically specific context.

Tangible User Intent

From a user experience perspective, park inquiries possess high transactional intent. Users asking

"Do I need a bear canister in Yosemite?" or "What are the fees for Banff?" require factual, unhallucinated answers backed by direct sources. This use case perfectly demonstrates the value of a RAG system over a base LLM. Base models frequently hallucinate specific fee structures or trail closures. By restricting the response parameters solely to the ingested canada-top-parks.json and us-top-parks.json datasets, the chatbot acts as a definitive, reliable guide.

3. DATAPIPELINE: SCRAPING METHODOLOGY

The foundation of any strong RAG system is the quality of its data. This project required extracting, cleaning, and formatting hundreds of pages of park documentation into machine-readable JSON files. The scraping pipeline was designed with validation checks to preserve data fidelity.

Methodological Approach

The scraping architecture used Python-based asynchronous requests together with BeautifulSoup for DOM parsing. The workflow followed four main steps:

1. **Target identification:** Sitemaps and index pages from Parks Canada and the US NPS were used to generate a queue of high-value URLs.
2. **DOM extraction:** The scraper parsed HTML to extract page titles, main content, and source URLs.
3. **Text sanitization:** Navigation, headers, footers, scripts, and styling were removed. Markdown conversion was used to preserve semantic structure, especially for tables.
4. **JSON structuring:** Cleaned content was stored in structured JSON arrays.

Technical Challenges Encountered

Table preservation: Standard text extraction often flattens HTML tables into unreadable strings. To address this, custom regular expressions and formatting logic were used to detect <table> elements and convert them into Markdown tables before appending them to plain text. This helps preserve row-column relationships for fees and seasonal data.

Rate limiting and stability: Exponential backoff and simulated human delays (time.sleep) were added to respect site limits and reduce request failures during ingestion.

4. Q&A GENERATION STRATEGY

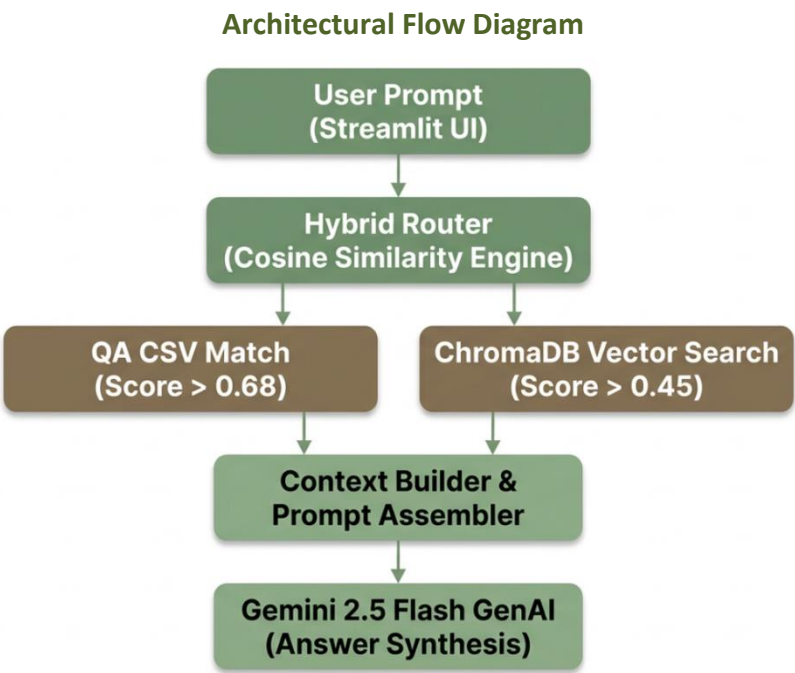
A defining feature of this architecture is the Hybrid retrieval approach, which relies on a pre-generated repository of Question/Answer pairs. This strategy acts as a high-speed caching layer for common, predictable queries, bypassing the computationally expensive dense vector search.

Quality Assurance of QA Pairs

After generation, the Q&A dataset was validated programmatically. Any pair that exceeded length thresholds or lacked a corresponding `source_url` mapping was removed. This ensures that when the hybrid router triggers a QA match, the answer is concise, accurate, and directly citable. The final dataset serves as a first-line defense against hallucination.

5. SYSTEM ARCHITECTURE & COMPONENT DESIGN

The system is designed as a modular, decoupled architecture, separating the ingestion pipeline, the vector storage, the retrieval logic, and the user interface. This ensures maintainability and allows independent scaling of the embedding models versus the generative LLM.



Core Components

Vector Database (ChromaDB): Selected for lightweight local execution, avoiding the need for cloud infrastructure during development. Documents were vectorized with sentence-transformers/all-MiniLM-L6-v2, which offers a good balance between efficiency and semantic performance for documentation-style content.

LLM Backend (Gemini 2.5 Flash): Chosen for low latency and a large context window. The Flash model works well for synthesizing responses from multiple retrieved chunks without losing important context.

6. HYBRID RETRIEVAL MECHANICS

The core innovation of the Park Ranger AI is the HybridRAGChatbot class logic. Standard vector search can sometimes struggle with highly specific factual queries (e.g., *"What is the exact price of a day pass?"*), returning semantically related text that lacks the precise number. The hybrid approach solves this.

Phase 1: The QA Matrix (Primary Retrieval)

When a user submits a query, it is embedded and compared against the questions in qa-combined-top-parks.csv. The system calculates cosine similarity. If the similarity score exceeds the QA_CONFIDENCE_THRESHOLD of **0.68**, the query is treated as a direct QA match. The verified answer and source metadata are then injected into the prompt.

Phase 2: Dense Vector Fallback

If the query is broad, conversational, or not a strong QA match, the system falls back to the ChromaDB parks_rag_v1 collection. Using the same query embedding, Chroma searches the chunked documentation. Retrieved chunks are included if they meet the DOC_CONFIDENCE_THRESHOLD of **0.45**. The system pulls the top 4 chunks (TOP_K_DOCS) and injects them into the final prompt.

Contextual Header Injection & Memory

To improve LLM comprehension, each document chunk is prefixed with its parent title during chunking. This contextual header injection helps the model keep park-specific information separated

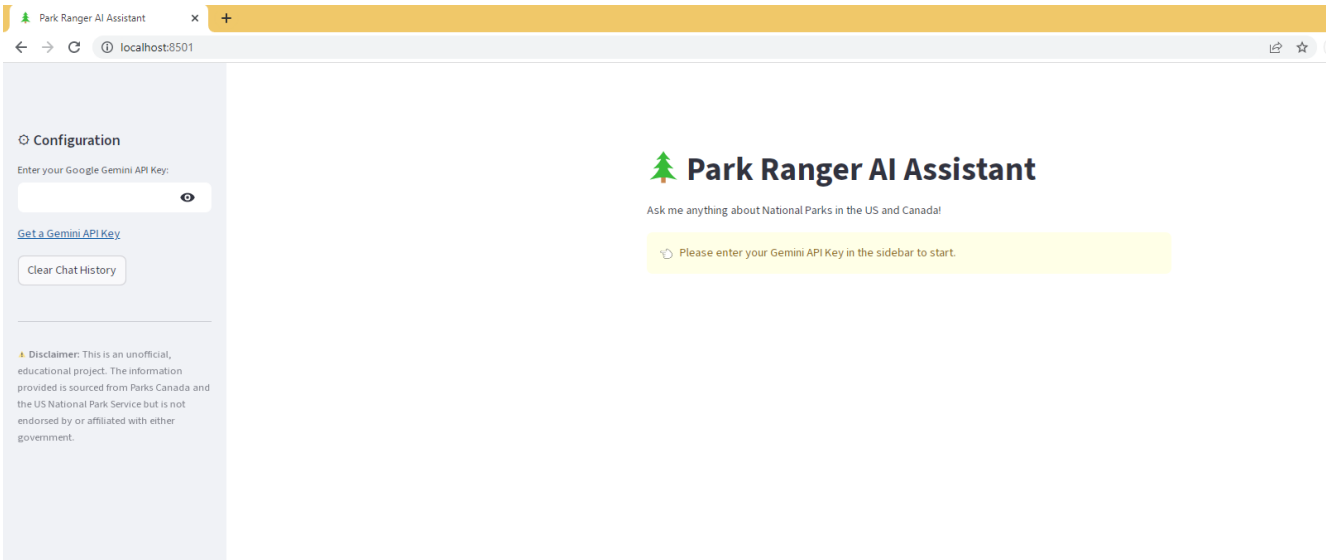
and reduces cross-contamination between parks.

The system also keeps the chat history in memory, allowing follow-up questions to be interpreted in context, such as “How much does it cost?”

7. UI/UX: CHATBOT INTERFACE DEPLOYMENT

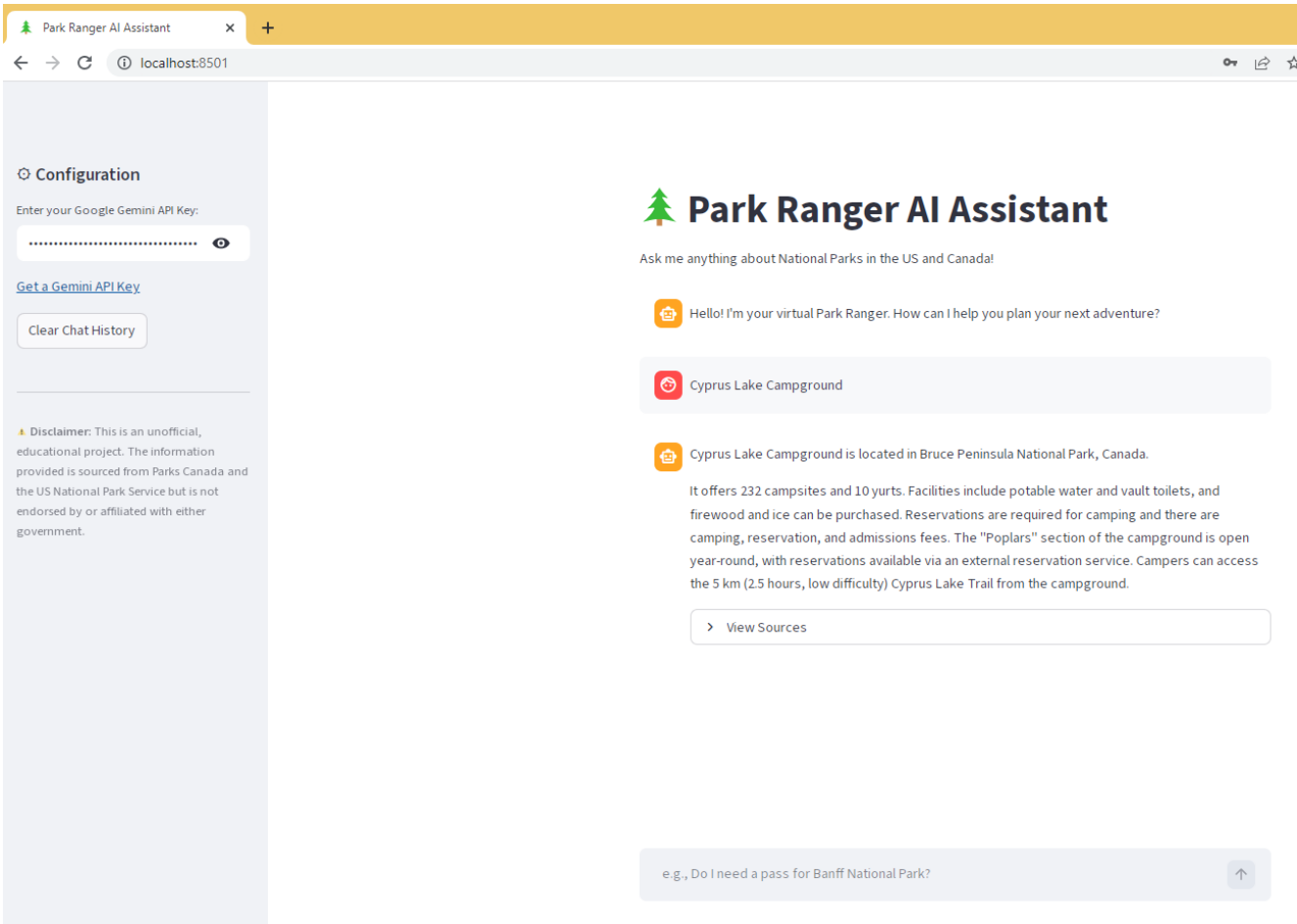
The application interface was deployed using Streamlit, which provides a fast Python-native way to build an interactive and stateful web app.

Application Interface Layout



Design Philosophy

The interface follows basic usability principles. API key authentication is handled in the sidebar, which prevents hardcoding credentials into the app. Chat history is preserved in Streamlit’s session_state, creating a smooth conversational flow. The “View Sources” expander adds transparency by showing the exact source URL and similarity score used to generate the response.



⊙ Configuration

Enter your Google Gemini API Key:

.....

[Get a Gemini API Key](#)

Clear Chat History

⚠ Disclaimer: This is an unofficial, educational project. The information provided is sourced from Parks Canada and the US National Park Service but is not endorsed by or affiliated with either government.

> View Sources



Bruce Peninsula National



Bruce Peninsula National Park, located in Canada on the Saugeen (Bruce) Peninsula, is a UNESCO World Biosphere Reserve, with the Niagara Escarpment forming its backbone.

For visitors, the park offers:

- **Activities:** Hiking (including part of the 782 km Bruce Trail, with trails like the Georgian Bay - Marr Lake Trail to Indian Head Cove and the Grotto, which is 3km, 3 hours, and varies from moderate to difficult), canoeing/kayaking, guided activities and interpretive programs, and winter activities.
- **Accommodations:** Front-country camping, yurts, back-country camping, and group camping.
- **Facilities:** A visitor centre and Parks Canada Shops.
- **Fees & Reservations:** Reservations are required for campsites, accommodations, and parking. There are various fees, and admission may be included with tour boat fees. The park also mentions "free admission and discounted camping fees, parking fees, and more," suggesting varying fee structures or special offers.
- **Planning:** Visitors can find maps, directions, hours of operation, and information on park regulations and safety.

⌵ View Sources

1. [Places to visit - Bruce Peninsula National Park](#) (Score: 0.8214)
2. [Bruce Peninsula National Park](#) (Score: 0.8017)
3. [Activities and experiences - Bruce Peninsula National Park](#) (Score: 0.7503)
4. [Hiking - Bruce Peninsula National Park](#) (Score: 0.7454)

e.g., Do I need a pass for Banff National Park?



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What is the highest peak in Yellowstone National Park



Eagle Peak, located in the southeastern part of Yellowstone National Park, is the highest peak at 11,358 feet.

> View Sources

e.g., Do I need a pass for Banff National Park?



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Park Ranger AI Assistant

Ask me anything about National Parks in the US and Canada!

History cleared!

 Hello! I'm your virtual Park Ranger. How can I help you plan your next adventure?

e.g., Do I need a pass for Banff National Park?

8. SYSTEM LIMITATIONS & QUALITY ASSURANCE ASSESSMENT

A realistic assessment of the system must acknowledge its constraints and identify where the current version may underperform.

Current Architecture Limitations

- **Static data decay:** The vector database is built from scraped JSON snapshots. Park fees, seasonal closures, and trail conditions change frequently, so without a scheduled re-scrape and embedding refresh, the knowledge base will become outdated.
- **Table understanding limits:** Even with Markdown preservation, complex nested tables are still flattened. The LLM may struggle with precise intersectional data, such as a specific row and column combination.

Out-of-Bounds Handling

The system is designed to fail gracefully. If both QA and document retrieval scores fall below the defined thresholds, the LLM receives an empty context string. The prompt then instructs the model to say: "I couldn't find enough support for that in the retrieved data. Please check the official Parks Canada or National Park Service sources for the latest details." This prevents the model from falling back on base knowledge and inventing potentially incorrect park information.

9. FUTURE IMPROVEMENTS & LEARNINGS

Architectural Iterations (V2 Roadmap)

Several upgrades could move this prototype toward a more robust production system:

- **Agentic routing:** Replace the current similarity-based router with an orchestrator agent that decides whether to query the vector DB, a structured SQL database for exact fees, or live web search for real-time updates.
- **Graph RAG:** Move beyond semantic similarity and use a knowledge graph to model entity relationships, such as park-to-trail or trail-to-difficulty links.
- **Cloud deployment:** Migrate ChromaDB to a managed vector store and deploy the Streamlit app in Docker on a cloud service to remove local dependency issues.

Key Learnings

This project reinforced a few important lessons. First, data quality matters more than model size. Considerable time spent preserving table structures and cleaning scraped content produced better results than simply using a larger LLM.

Second, resilient applications need retry logic. Moving from prototype scripts to a Streamlit app required handling temporary Gemini 503 errors, and exponential backoff was essential for maintaining a stable user experience.

Ultimately, this deployment shows that effective RAG depends not just on model output, but on disciplined data engineering, robust error handling, and clear system design.